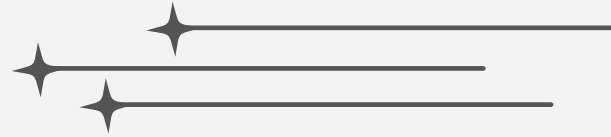


DATA ANALYTICS PORTFOLIO PROJECT

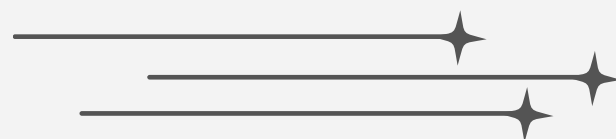


Retail Sales, Store Performance, Department Analysis,
Seasonality

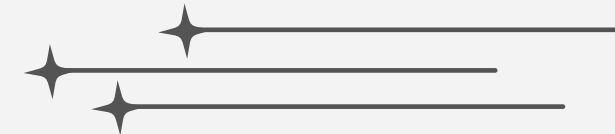
Walmart Retail Sales Analysis

Excel • SQL • Python • Power BI

Presented By: **Md Avesh**



Executive Summary



Walmart Retail Sales Analysis is an end-to-end data analytics project focused on understanding sales performance across stores, departments, store types, and time periods.

The project transforms raw retail transaction data into actionable business insights using Excel, SQL, Python, and Power BI.

The analysis focuses on identifying high- and low-performing stores and departments, evaluating store size and sales relationships, analyzing monthly and seasonal trends, and comparing holiday versus non-holiday sales performance.

The final outcome is an interactive 4-page Power BI dashboard designed to support business performance monitoring and data-driven decision-making.

421K+

Sales Records

45

Stores

81

Departments

\$6.74B

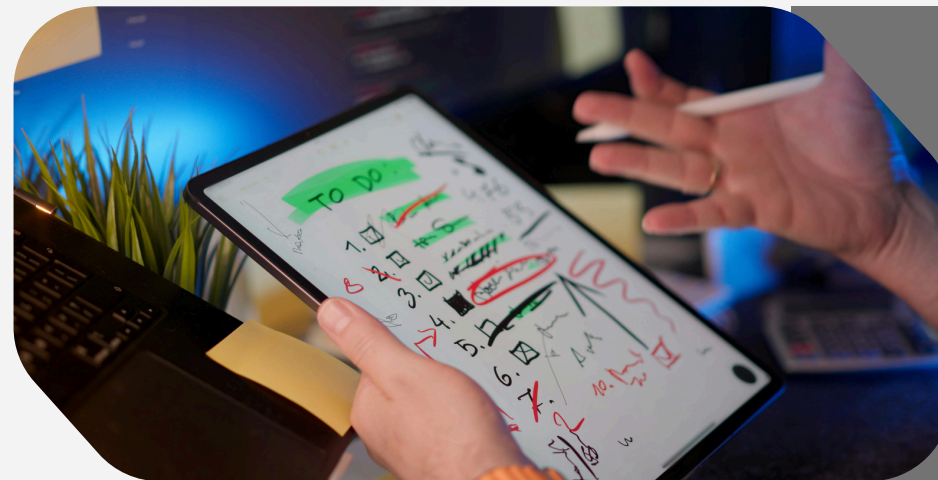
Total Sales



How can Walmart turn large-scale retail sales data into clear, actionable business insights?

01 — Store Performance

— Identify high- and low-performing stores and understand performance variation across locations.



02 — Department Performance

— Evaluate department-level sales contribution and identify strong and weak departments.

03 — Sales Trends & Seasonality

— Understand monthly sales patterns and changes in performance over time.

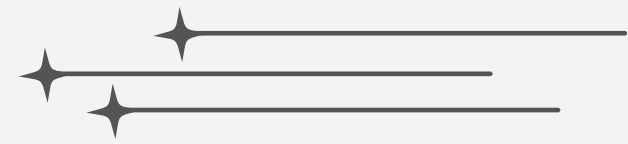
04 — Holiday Impact

— Compare holiday and non-holiday sales performance to understand seasonal demand patterns.

Business Problem

Build a reliable analytical workflow that converts raw sales data into decision-ready insights.





Objective: Convert raw retail data into meaningful insights that support better business decisions.

Project Objectives



01 — Measure Sales Performance

Analyze overall sales performance across stores, departments, store types, and time periods.

02 — Identify Performance Leaders

Identify top-performing stores and departments based on sales contribution.

03 — Analyze Sales Drivers

Examine relationships between store size, store type, and sales performance.

04 — Understand Seasonality

Analyze monthly trends and compare holiday versus non-holiday sales behavior.

05 — Enable Data-Driven Decisions

Transform analysis into an interactive Power BI dashboard for business performance monitoring.



Dataset Overview

Data Scope: Historical weekly sales data covering multiple stores and departments, enabling store-level, department-level and time-based performance analysis.

**421,570
Records**

**45
Stores**

**81
Departments**

**2010–2012
Sales Period**



Tools & Technologies

Raw Data → Excel → SQL → Python → Power BI → Business Insights

01 — Excel

Data cleaning, validation, transformation and exploratory analysis.

02 — SQL

Data querying, aggregation and business-oriented analysis.

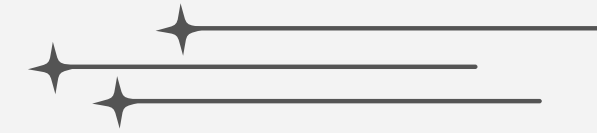
03 — Python

Store-level analysis, calculations, ranking and analytical validation.

04 — Power BI

Interactive dashboard, KPIs, trends, comparisons and business reporting.





Data Preparation & Cleaning

01 — Raw Data Source Files

- train.csv
- test.csv
- features.csv
- stores.csv

Initial checks:

- Row & column counts
- Data types
- Missing values
- Duplicate records
- Date consistency
- Sales anomalies

02 — Cleaning & Transformation

Data Standardization

Standardized date fields and data types.

Missing Value Handling

Reviewed missing values and removed CPI & Unemployment from the cleaned analytical features due to substantial missing data.

Data Integration

Combined sales, store and relevant feature information into a unified analytical dataset.

Data Quality Checks

Reviewed negative/zero sales records and validated key metrics.

03 — Final Analytical Dataset

Walmart Master Dataset
421,570 rows × **14** columns

Integrated dataset used for downstream SQL, Python and Power BI analysis.



Raw Files → Clean → Transform → Integrate → Validate → Analysis

SQL Analysis



01 – Sales Aggregation

Calculated total and average sales across stores, departments and time periods.

02 – Store Performance

Compared store-level sales performance and identified top and bottom performers.

03 – Department Analysis

Evaluated department-level sales contribution and performance differences.

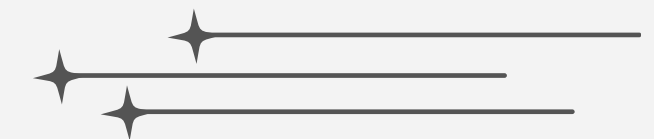
04 – Business Filtering

Used filtering, grouping and aggregation to answer business-focused questions.

SQL Visual

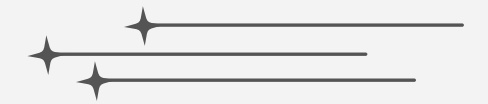
```
SELECT  
    Store,  
    SUM(Weekly_Sales) AS Total_Sales  
FROM Walmart_Sales  
GROUP BY Store  
ORDER BY Total_Sales DESC;
```

Example: Store-level sales ranking





Python Analysis



01 – Data Validation

Validated cleaned data and key analytical metrics.

02 – Store-Level Analysis

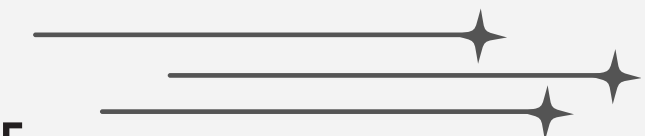
Calculated store-level record counts, average sales and total sales.

03 – Performance Ranking

Ranked stores based on total sales to identify performance leaders.

Python Code Card

```
store_analysis = (  
    df.groupby(["Store", "Type", "Size"])  
        .agg(  
            record_count=("Weekly_Sales", "count"),  
            average_sales=("Weekly_Sales", "mean"),  
            total_sales=("Weekly_Sales", "sum")  
        )  
        .reset_index()  
    )
```



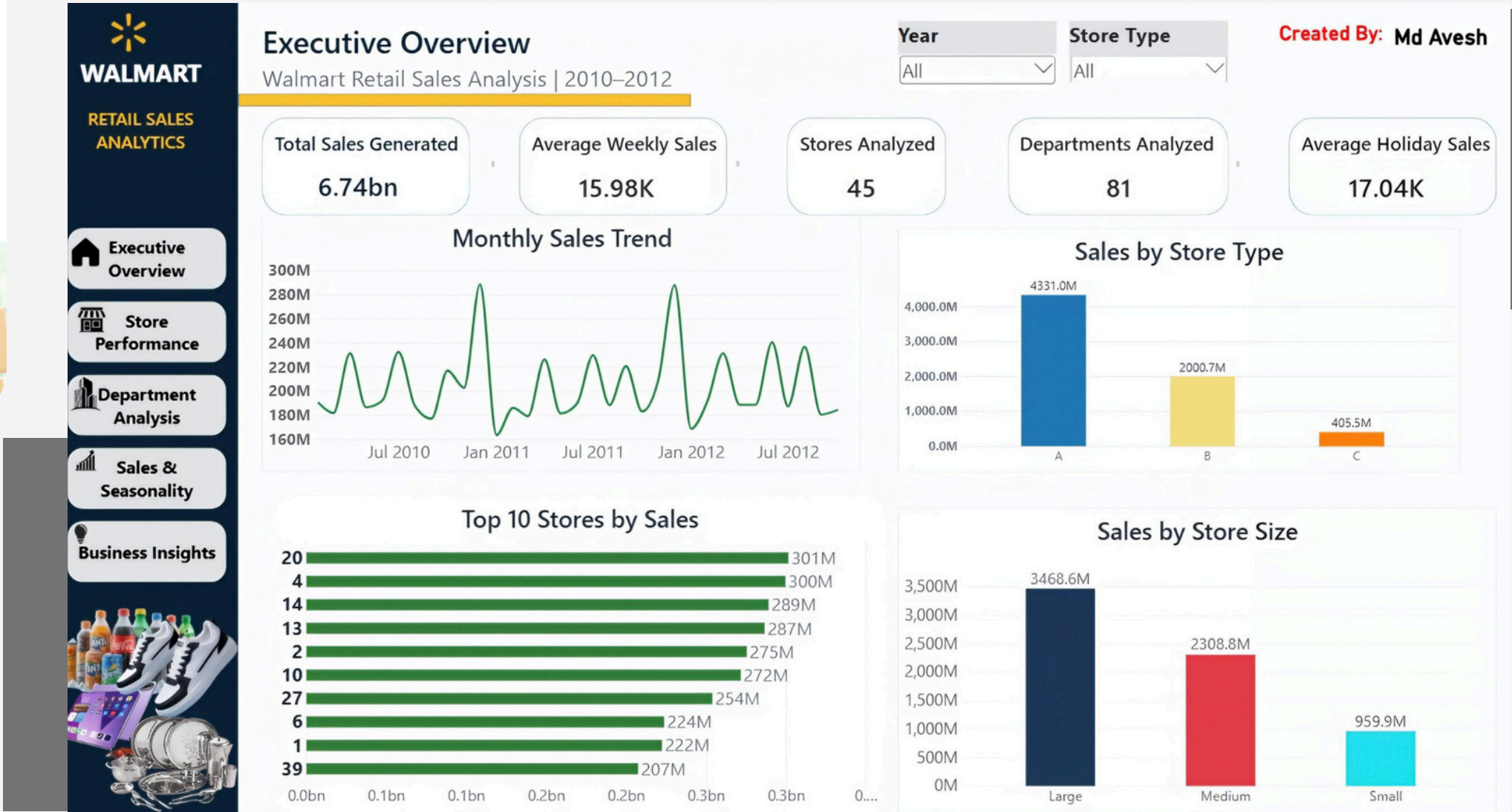
04 – Analytical Output

Created a structured store-level analysis dataset for Power BI reporting.



Power BI Dashboard

Interactive Sales Intelligence Dashboard



Sales Overview

Track total sales and weekly sales performance.

Store Performance

Compare store-level performance and rankings.

Department Analysis

Evaluate department contribution and performance.

Seasonality

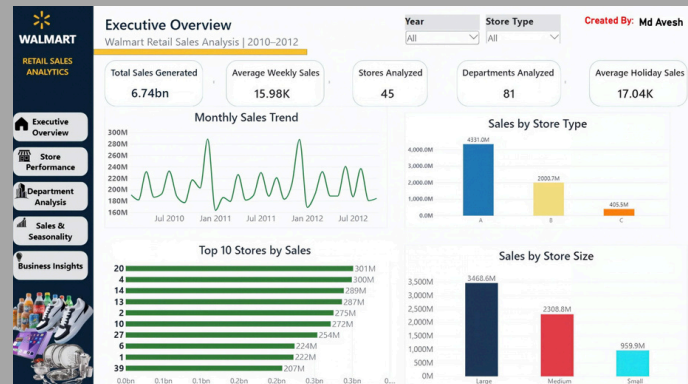
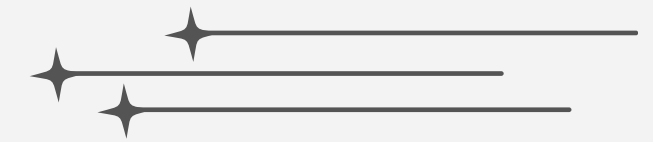
Analyze monthly trends and holiday sales behavior.

Excel → SQL → Python → Power BI → Business Insights



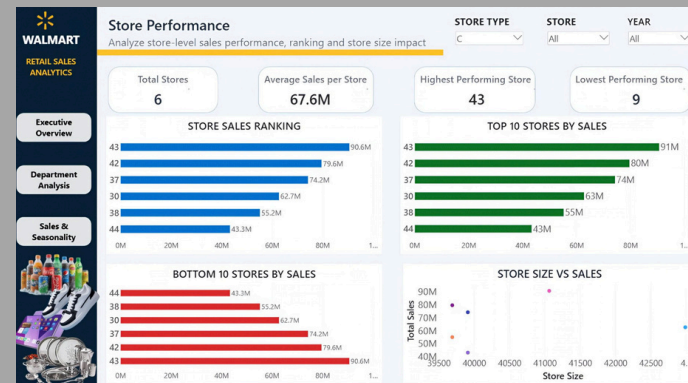


Dashboard Pages



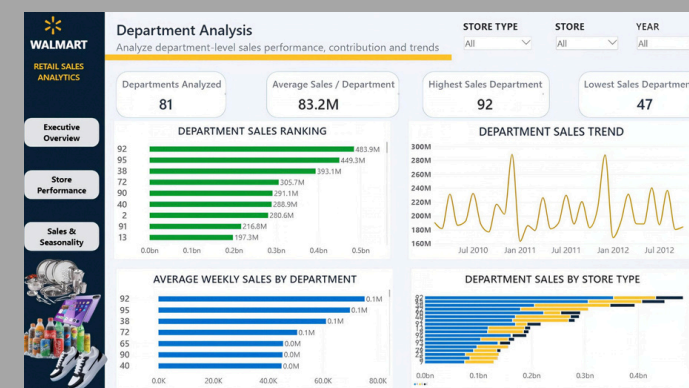
01 – Executive Overview

Overall sales performance, KPIs, store types, top stores and store-size analysis.



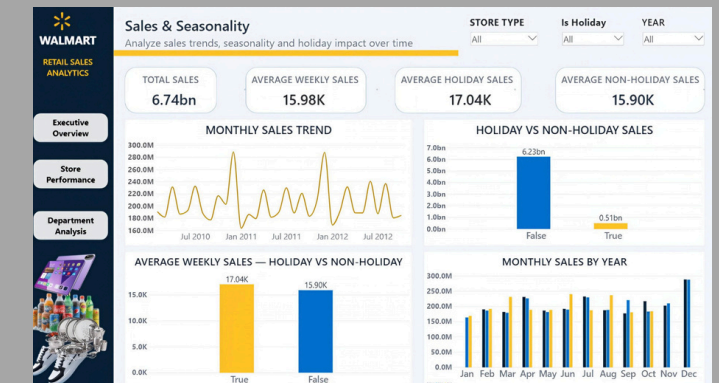
02 – Store Performance

Store rankings, top/bottom performers and store size vs sales relationship.



03 – Department Analysis

Department rankings, sales trends, average weekly sales and store-type contribution.



04 – Sales & Seasonality

Monthly sales trends, holiday comparison and yearly seasonality.





Key Performance Indicators

Holiday vs Non-Holiday

Holiday weeks recorded a higher average weekly sales level than non-holiday weeks.

Holiday: ~ \$17.04K

Non-Holiday: ~ \$15.90K



KPI Cards

01 – Total Sales
\$6.74B

Overall sales generated across the analyzed dataset.

02 – Average Weekly Sales
\$15.98K

Average weekly sales across sales records.

03 – Stores Analyzed
45

Total stores included in the analysis.

04 – Departments Analyzed
81

Total departments covered.

05 – Average Holiday Sales
\$17.04K

→ Average weekly sales during holiday weeks.

Key Business Insights



01 – Store Performance Varies Significantly

Sales performance differs considerably across stores, creating clear high- and low-performing groups.

02 – Larger Stores Generate Higher Sales

Larger stores generally show higher total sales, indicating a positive relationship between store size and sales performance.

03 – Sales Show Clear Seasonal Patterns

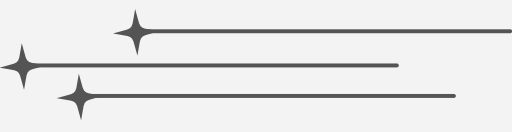
Monthly sales vary throughout the year, with noticeable differences across months and years.

04 – Holiday Weeks Show Higher Average Sales

Sales performance differs considerably across stores, creating clear high- and low-performing groups.

05 – Negative Sales Require Monitoring

The dataset contains **1,285 negative sales records**, which should be investigated as potential data-quality or transaction-level exceptions.



Business Recommendations

05 – Monitor Sales Exceptions

Investigate negative and zero-sales records to distinguish genuine business transactions from potential data-quality issues.



01 – Focus on Underperforming Stores

Investigate low-performing stores and identify operational or market-specific factors affecting sales.

02 – Benchmark High-Performing Stores

Study the practices and characteristics of top-performing stores to identify transferable best practices.

03 – Plan Around Seasonal Demand

Use historical monthly patterns to improve inventory, staffing and promotional planning during high-demand periods.

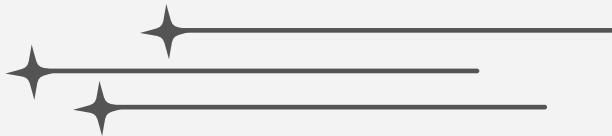
04 – Strengthen Holiday Planning

Use holiday-period sales patterns to support promotional, inventory and workforce planning.

Recommendation: Use the dashboard as a recurring performance-monitoring tool rather than a one-time analysis.



Project Outcome



➔ **Transformed 421,570 retail sales records into an interactive business intelligence solution.**

01 – Unified Analysis

Integrated store, department and sales information into a structured analytical workflow.

02 – Performance Visibility

Enabled clear comparison of store and department performance.

03 – Trend Understanding

Revealed monthly, yearly and holiday-related sales patterns.

04 – Decision Support

Created an interactive Power BI dashboard for performance monitoring and business analysis.



Final Deliverable: A professional 4-page Power BI dashboard supported by Excel, SQL and Python analysis.



Skills Demonstrated

01 – Data Preparation

- Excel
- Power Query
- Data Cleaning
- Data Validation

02 – Data Analysis

- SQL
- Aggregation
- Filtering
- Business Analysis

03 – Programming & Analytics

- Python
- Pandas
- Data Aggregation
- Ranking Analysis

04 – Business Intelligence

- Power BI
- DAX
- Data Modeling
- Interactive-Dashboards



Core Capability: Transforming raw business data into structured analysis, interactive dashboards and actionable insights.



Final Portfolio Summary

Walmart Retail Sales Analysis

An end-to-end data analytics project focused on turning retail data into business insights.

DATA

421,570 records
45 stores
81 departments

ANALYSIS

Excel • SQL • Python
Cleaning • Analysis • Ranking
• Validation

BI SOLUTION

Power BI
4 Interactive Dashboard Pages
KPIs • Trends • Performance Analysis

From raw retail data to actionable business intelligence.